

Mechanical Power Module Consortium (MPMC)

The Global Standards Body for Universal Mechanical Power Interfaces

The Mechanical Power Module Consortium is the international standards organization responsible for defining, governing, and advancing the Mechanical Power Module (MPM) — a universal, interoperable mechanical interface designed to modernize global manufacturing, robotics, energy systems, and distributed industrialization.

MPM establishes a single, sovereign mechanical power primitive, enabling machines, tools, vehicles, and modular infrastructure to connect, operate, and exchange mechanical power with the same simplicity and reliability that USB brought to digital systems.

Our Mission

To create and maintain a universal mechanical interface standard that:

Expands global manufacturing capability

Enables interoperable mechanical systems

Reduces cost and complexity across industries

Supports distributed industrialization in emerging regions

Ensures safety, reliability, and sovereign access to mechanical power

The MPM Standard is designed for global adoption, open licensing, and long-term stability.

The MPM Standard

The Mechanical Power Module (MPM) defines:

Core Geometry — Universal mounting rail, torque bus, alignment features

Mechanical Interface — Standardized torque, rotational power, and load transfer

Safety Envelope — Catastrophic shear, runaway torque, thermal limits

Thermal Envelope — Passive and active dissipation provisions

Module Classes — Power heads, driven modules, hybrid modules

Compliance Test Suite — Certification, conformance, and interoperability testing
MPM v1.0 is currently in pre-ratification stabilization.

Governance Structure

The Consortium operates through a three-body governance model:

Standards Board

Technical authority over the MPM specification, revisions, and ratification. Chaired by the Founding Technical Architect.

Compliance Authority

Certification, conformance testing, audit protocols, and dispute resolution.

Deployment Council

Regional adoption, manufacturing alignment, and global rollout.

Learn more:

Governance Bylaws

Standards Board Operating Rules

Compliance Authority

Deployment Council

Global Participation

The Consortium is forming a founding cohort of organizations across:

Africa (regional blocs)

Latin America (industrial clusters)

United States (national lab + private sector)

Europe (manufacturing and robotics firms)

Asia (emerging industrial partners)

Membership categories include:

Founding Members

Voting Members

Non-Voting Members

Technical Contributors

Learn more: [Membership Overview](#)

Why MPM Matters

Mechanical power today is fragmented across incompatible interfaces, proprietary systems, and region-specific designs. MPM provides:

A universal mechanical standard

A shared global manufacturing language

A safe, interoperable mechanical ecosystem

A foundation for distributed industrialization

A pathway for modular robotics and infrastructure

MPM is to mechanical power what ISO containers were to logistics — a unifying, transformative standard.

Shipping Container Dimensions — R R Plett Trucking

Iso Container Standards Understand Shipping Container Sizes

What Is Interface Diagram at Ruby Lay blog

DARPA selects Intact Solutions to Advance Digital Twins for Resilient ...

Current Status (October 2026)

Charter v1.0 in final review

Governance stack nearing completion

MPM Standard v1.0 in stabilization

Founding Assembly Session #1 scheduled

Membership negotiations active with 11 organizations

Reference implementation architecture in development

Follow updates: Consortium News

Get Involved

Organizations interested in participating in the development, adoption, or deployment of the MPM Standard may request:

Membership information

Technical contributor access

Early access to draft specifications

Pilot program participation

Regional deployment